Interaction of droplets with vortical structures: to understand atmospheric clouds

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Over the last decade, considerable progress has been made in understanding cloud formation and rain initiation [1]. However, the development of cloud droplets into rain droplets, which initiate precipitation in warm clouds, is still poorly understood. The rapid development of cloud droplets into rain droplets is believed to be caused by collision and coalescence, which is significantly governed by the complex interaction of background turbulent flow structures with cloud droplets[2]. A good approximation of this complex interaction is the interaction of cloud-like droplets with a well-established vortical flow field (Gaussian vortex: figure 1(b)), which represents the eddies in turbulent flow [3]. Therefore, in the present study, we experimentally investigate the interaction of cloud-like droplets with the vortical structures to better understand cloud microphysics, as shown in figure 1. Flow diagnostic techniques, such as shadowgraphy, multiphase time-resolved PIV, and PTV, are utilized to study the complex dynamics of clustering and collision in clouds.

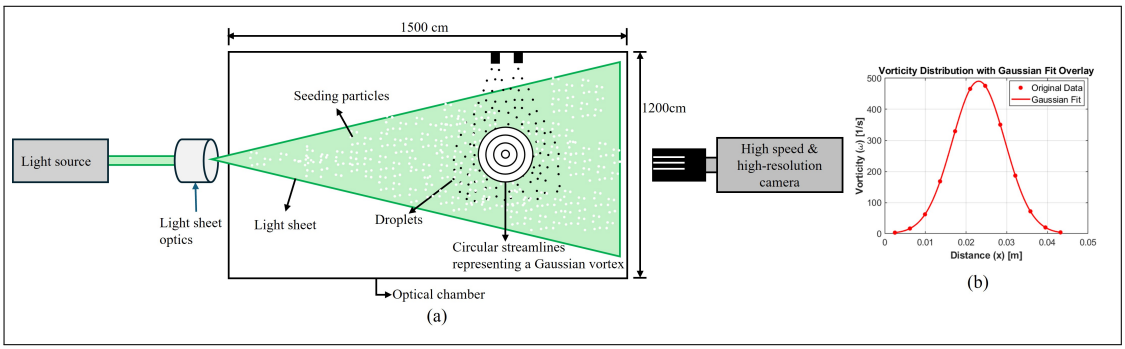

Figure 1: (a) Schematic diagram of the experimental facility; (b) vorticity distribution along the vortex.

We have generated cloud-like droplets with size distribution and number density similar to those in atmospheric clouds. We utilized a high-speed shadowgraphy technique to study the interaction of droplets with vortical structure. After the interaction, one can expect a preferential concentration of droplets where droplets will migrate from high to low vorticity regions, generating voids and clusters, and we utilized Voronoi analysis to identify them. A typical Voronoi diagram of the droplets after the interaction is shown in figure 2(a), where the clusters and voids are classified by comparing the pdfs of experimental normalized Voronoi areas with the random Poisson process (RPP) distribution as shown in figure 2(b). A detailed explanation of this interaction and the effect of various non-dimensional parameters on cluster and void formation will be discussed during the presentation.

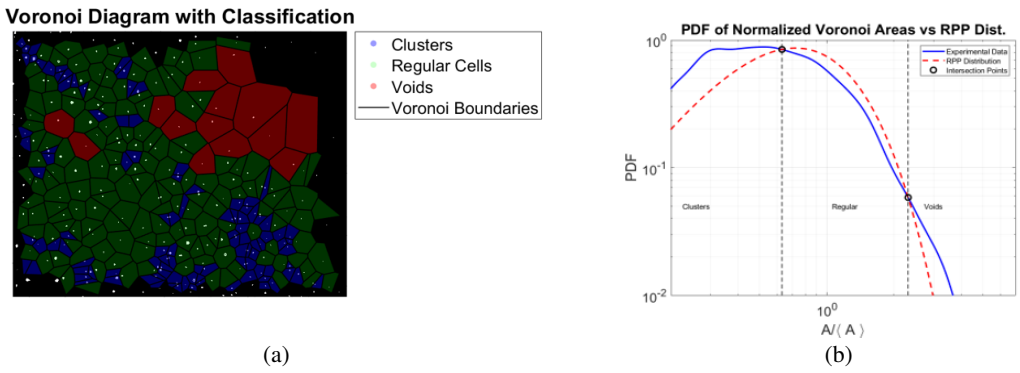

Figure 2: (a) Voronoi diagram showing the clusters and voids generated after the vortex influences the droplets; (b) PDF comparision of experimental normalised voronoi area with RPP distribution.

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[1] Shaw, *Annual Review of Fluid Mechanics* **35**, 183–227 (2003)
[2] Grabowski & Wang, *Annual Review of Fluid Mechanics* **45**, 293–324 (2013)
[3] Linden, *Journal of Fluid Mechanics* **60**, 467–480 (1973)